1 Solution — GIAM 1.2 Problem 6

1.1 Problem

Find a counterexample for the conjecture that $x^2 - 31x + 257$ evaluates to a prime number whenever x is a natural number.

1.2 The Counterexample

Write $f(x) = x^2 - 31x + 257$. The smallest natural-number counterexample is

$$\boxed{x = 32 \ : \quad f(32) \ = \ 32^2 - 31 \cdot 32 + 257 \ = \ 1024 - 992 + 257 \ = \ 289 \ = \ 17^2.}$$

Since $289 = 17 \times 17$ is composite, the conjecture is false. ■

1.3 It Holds for a Long Time First

What makes this conjecture so seductive is that $f(x)$ really *is* prime for every natural number from 1 all the way to 31 — thirty-one primes in a row — before it finally breaks at 32:

x	$f(x)$	**prime?**	x	$f(x)$	**prime?**
1	227	yes	9	59	yes
2	199	yes	10	47	yes
3	173	yes	11	37	yes
4	149	yes	12	29	yes
5	127	yes	13	23	yes
6	107	yes	14	19	yes

7	89	yes	15	17	yes
8	73	yes	16	17	yes

Table 1.1

For $x = 17, 18, \ldots, 31$ the values *mirror* those above (see the symmetry remark) and are likewise all prime — $f(31) = 257$. Then:

x	$f(x)$	**prime?**
32	289	**no** — $289 = 17^2$
33	323	**no** — $323 = 17 \cdot 19$
34	359	yes (prime again!)

Table 1.2

So $x = 32$ is the *first* counterexample, and the failures do not even persist — $f(34) = 359$ is prime again.

1.4 The Shape of the Values

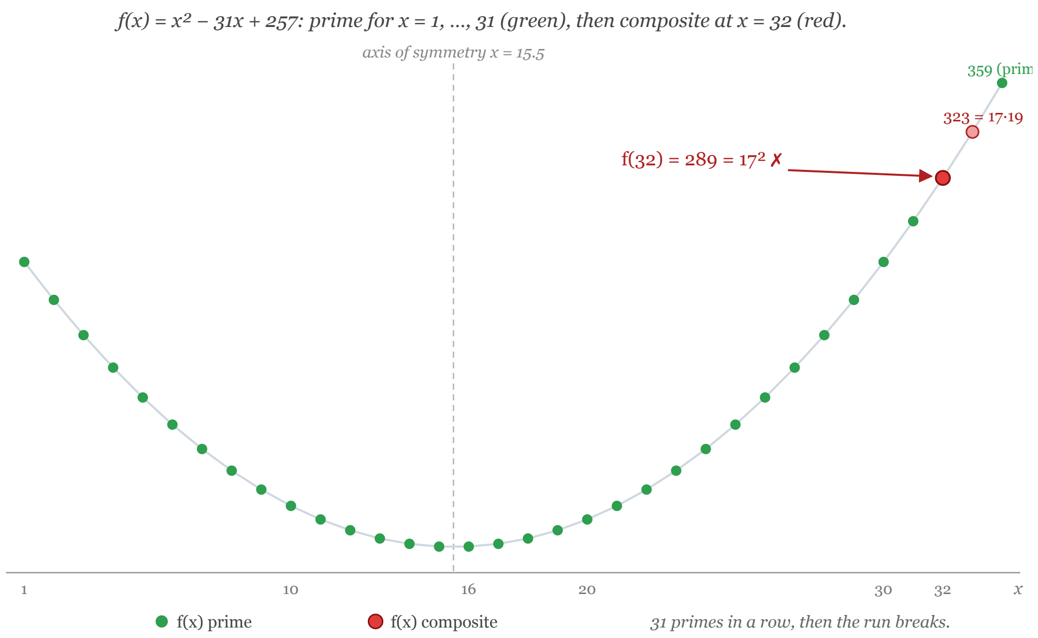

Figure 1.1: The parabola f(x) = $x^2 - 31x + 257$, symmetric about x = 15.5. Green dots (x = 1..31) are prime; the value bottoms out at 17 near the vertex. Red dots at x = 32 ($289 = 17^2$) and x = 33 ($323 = 17 \cdot 19$) are the first composites — then x = 34 is prime again.

The plot shows the parabola dipping to its minimum value 17 at the vertex ($x = 15$ and 16) and climbing symmetrically on either side. Every integer dot from $x = 1$ to 31 is green (prime); the first red dot is $x = 32$.

1.5 Remark — The Symmetry $f(x) = f(31 - x)$

The repeated values in the table are no accident. A parabola is symmetric about its vertex, which for $f(x) = x^2 - 31x + 257$ sits at $x = \frac{31}{2} = 15.5$. Hence

$$f(x) \;=\; f(31 - x) \qquad \text{for every } x.$$

So $f(1) = f(30) = 227$, $f(2) = f(29) = 199$, and so on — the values for $x = 16, \ldots, 31$ are a mirror image of those for $x = 15, \ldots, 0$. This is why checking $x = 1, \ldots, 16$ already determines everything up to $x = 31$.

1.6 Remark — This Is Euler's "Lucky" Polynomial in Disguise

Why is f prime for such a long run? Shift the variable by completing the square. Setting $y = x - 16$:

$$f(x) \;=\; (y + 16)^2 - 31(y + 16) + 257 \;=\; y^2 + y + 17.$$

The polynomial $y^2 + y + 17$ is one of **Euler's prime-generating polynomials**: it is prime for every $y = 0, 1, 2, \ldots, 15$, and by the same symmetry ($y^2 + y + 17 = (-y - 1)^2 + (-y - 1) + 17$) also for $y = -1, -2, \ldots, -16$. That is a run of 32 consecutive prime values, $y \in \{-16, \ldots, 15\}$ — exactly our $x \in \{0, \ldots, 31\}$.

The most famous member of this family is **Euler's** $y^2 + y + 41$, which is prime for $y = 0, \ldots, 39$ (forty values).

1.7 Remark — The Deep Reason: a Heegner Number

The run-length is governed by a remarkable fact from algebraic number theory. The polynomial $y^2 + y + q$ is prime for all $y = 0, 1, \ldots, q-2$ **if and only if** the imaginary quadratic field $\mathbb{Q}(\sqrt{1-4q})$ has *class number one*. For $q = 17$:

$$1 - 4q \;=\; 1 - 68 \;=\; -67,$$

and 67 is one of the nine **Heegner numbers** $(1, 2, 3, 7, 11, 19, 43, 67, 163)$ — the discriminants for which class number one holds. So $y^2 + y + 17$ achieving its maximal prime run is a shadow of $\mathbb{Q}(\sqrt{-67})$ being a unique-factorization domain. Euler's $y^2 + y + 41$ corresponds to the largest Heegner number, 163 (since $1 - 4 \cdot 41 = -163$) — which is also why $e^{\pi\sqrt{163}}$ is famously almost an integer.

The conjecture in this problem was, in a sense, "true for a deep reason" right up until $y = 16$ (i.e. $x = 32$) — the first value *past* the run that the class-number-one phenomenon protects.

1.8 Remark — An Infinite Family of Counterexamples

Once one counterexample is found, infinitely many follow. Reducing f modulo 17 using the shifted form $f(x) = (x-16)^2 + (x-16) + 17$:

$$f(x) \;\equiv\; (x-16)^2 + (x-16) \;=\; (x-16)(x-15) \pmod{17}.$$

This is $\equiv 0 \pmod{17}$ whenever $x \equiv 15$ or $x \equiv 16 \pmod{17}$. So 17 divides $f(x)$ for

$$x \;=\; 15, 16, \quad 32, 33, \quad 49, 50, \quad 66, 67, \; \ldots$$

At $x = 15, 16$ the value *equals* 17 (still prime), but for every larger member of this list $f(x) > 17$ while remaining divisible by 17 — hence composite. So $x = 32, 33, 49, 50, 66, 67, \ldots$ are *all* counterexamples. The conjecture fails infinitely often.

1.9 Remark — No Nonconstant Polynomial Is Always Prime

The deepest moral is that this conjecture was *doomed from the start*: **no nonconstant polynomial with integer coefficients can produce a prime for every natural-number input.** Here is the short proof.

Let f be such a polynomial and suppose, for contradiction, that $f(n)$ is prime for all $n \geq 1$. Let $p = f(1)$, a prime. For any integer k, expand $f(1+kp)$: every term beyond the constant carries a factor of $1+kp$, so

$$f(1+kp) \;\equiv\; f(1) \;=\; p \;\equiv\; 0 \pmod{p}.$$

Thus $p \mid f(1+kp)$ for every k. If each such value is prime, it must *equal* p (the only prime divisible by p). But then the polynomial $f(x) - p$ would have infinitely many roots $x = 1, 1+p, 1+2p, \ldots$ — impossible for a nonconstant polynomial. Contradiction. ■

So *every* conjecture of the form "this polynomial is always prime" is false; the only question is *which* input is the first counterexample. For $f(x) = x^2 - 31x + 257$, that input is $x = 32$. (Applied directly, the proof would point to $x = 1 + 227k$ as a guaranteed counterexample — e.g. $f(228)$ is a large multiple of 227 — but the genuine *smallest* counterexample, 32, is found by direct search.)

1.10 Remark — The Echo of Problem 5

This is the second false conjecture in a row (after "$2^p - 1$ is prime for prime p" in Problem 5), and the lesson is identical: **a pattern that holds for many cases — even thirty-one in a row — is not a proof.** Both conjectures were specifically chosen by the textbook to fail, precisely to inoculate against the temptation to believe a statement just because the first several cases check out. Verifying $f(1), \ldots, f(31)$ feels overwhelmingly convincing, yet $f(32)$ quietly demolishes it.